UNIT 6

RATIO AND PROPORTION

RATIO

Comparison of two quantities of the same kind is called ratio. This comparison is always in the form of numbers. A ratio is represented by a fraction or the symbol " : ". If the cost of two toffees is Rs 2 and Rs 3 then their ratio will be or 2:3, read as the "ratio of 2 is to 3".

A ratio has no units.

The order in which a ratio is written is important.

To find ratio between two quantities it is necessary that they must be of the same kind.

Know that of the two quantities forming a ratio, the first one is called antecedent and the second one consequent:

For Example:

Ratio	Antecedent	Consequent
2 : 3	2	3
5 : 7	5	7
18 : 23	18	23

Calculate ratio of two numbers:

Example 1: Find the ratio of the following:

(i) Ali has Rs 500 and Seema has Rs 309. Write the ratio of their amounts.
Solution: The required ratio is 500:309.

(ii) Nazir got 35 litres of fuel filled in his car and Aijaz filled 48 litres of fuel in his car. Write the ratio of quantities of their fuel.
Solution: The required ratio is 35:48.

Reduce given ratio into lowest (equivalent) form:

Example 2: Find equivalent ratios of 40:8.

Solution:

40 : 8
= 40 ÷ 2 : 8 ÷ 2 (Dividing by 2)
= 20 : 4

or 40 : 8
= 40 ÷ 4 : 8 ÷ 4 (Dividing by 4)
= 10 : 2

or 40 : 8
= 40 ÷ 8 : 8 ÷ 8 (Dividing by 8)
= 5 : 1

So, 20 : 4, 10 : 2, and 5 : 1 are some of the lower equivalent ratios of 40 : 8.

Note: In the above example, 5 : 1 is **the lowest equivalent form** of 40 : 8.

Example 3: Find the ratio between Rs 210 and Rs 105 and write in lowest equivalent form.

Solution:

= 210 : 105
= 70 : 35 (Dividing both by 3)
= 14 : 7 (Dividing both by 5)
= 2 : 1 (Dividing both by 7)

Example 4: Find the ratios of the following quantities and write in lowest form.

Solution:

(i) In the given ratio 15 kg : 700 g, the units are different.
As 1 kg = 1000 g
Therefore, 15 kg = (15 × 1000) = 15000 g

Hence, the ratio in the lowest form

= 15 kg : 700 g

= 15000 g : 700 g

= 15000 : 700

= 150 : 7 (dividing both by 100)

Relationship between ratio and fraction:

Example 1: Two pieces of cloth are of 2 m 50 cm and 75 cm length respectively. Find the ratio between their lengths.

Solution:

Let us convert the given lengths in the same units.

As, 1 m = 100 cm,

So, 2 m 50 cm = (2 × 100) cm + 50 cm

= (200 + 50) cm = 250 cm

Hence, the ratio of their lengths = 2 m 50 cm : 75 cm

= 250 cm : 75 cm

$= \frac{\cancel{250}^{10}}{\cancel{75}_{3}}$

$= \frac{10}{3} = 10 : 3$

Example 2: Find ratio of the following and write in lowest form.

(i) $\frac{1}{5}$ and $\frac{2}{3}$ **(ii)** $1\frac{3}{5}$ m and $\frac{7}{10}$m

(iii) 1.5 and $7\frac{1}{2}$ **(iv)** Rs. 3.45 and Rs. 6

Solution:

(i) $\frac{1}{5} : \frac{2}{3} = \frac{1}{5} \times 15 : \frac{2}{3} \times 15$

(Multiplying both the LCM of 5 and 3)

= 3 : 10

(ii) $1\frac{3}{5} : \frac{7}{10} = \frac{8}{5} : \frac{7}{10}$

$= \frac{8}{5} \times 10 : \frac{7}{10} \times 10$

(Multiplying both the LCM of 5 and 10)

$= 16 : 7$

(iii) $1.5 : 7\frac{1}{2} = 1.5 : \frac{15}{2}$

$= 1.5 \times 10 : \frac{15}{\cancel{2}_1} \times \cancel{10}^5$ (Multiplying both by 10)

$= 15 : 15 \times 5$

$= 1 : 5$ (dividing both by 50)

(iv) 3.45: 6 = 345: 600

(As 3.45 is two degree decimal, so we convert it into a whole number by multiplying with 100.)

= 69: 120 (Dividing both by 5)

= 23: 40 (Dividing both by 3)

EXERCISE 6.1

Q1. Reduce the following into lowest equivalent form.

(i) 4 : 50 **(ii)** 0.8 : 72 **(iii)** 3.5 : 4.9

(iv) $\frac{13}{60} : \frac{7}{15}$ **(v)** $\frac{5}{6} : \frac{3}{10}$ **(vi)** $2 : \frac{5}{3}$

Q2. Find ratio of the following and write in lowest form.

(i) Rs 150 and Rs 180 **(ii)** 250 cm and 1 m

(iii) 700 g and 2 kg **(iv)** 3 hours and 210 minutes

(v) 5 years and 3 months

Q3. Represent the following into fractions.

(i) 1 : 5 **(ii)** 2 : 19 **(iii)** 8 : 1 **(iv)** 75 : 76 **(v)** x : y

Q4. In a Science test, 25 students out of 45 students of class VI were passed. Find the ratio between the passed students and total students.

Q5. Arshad earns Rs 20,000 per month. Find the ratio of his monthly income and expenditure if he saves Rs. 5000 per month.

Q6. The measures of sides of two squares are 2 cm and 5 cm. Find the ratio of their perimeters.

Q7. Weight of a sack of flour is 16 kg and the weight of another sack is 14kg 400g. Find the ratio of their weights.

PROPORTION

Two equivalent ratios form a proportion.

For example: 3:4 = 6:8 is a proportion.

Read as: 3 is to 4 as 6 is to 8.

The two middle elements of a proportion are called **means**, where as the elements at both ends of a proportion are known as **extremes**. For example:

$$\underbrace{2 : \overbrace{7 :: 6}^{\textbf{Means}} : 21}_{\textbf{Extremes}}$$

Product of extremes = product of means.

Example 1: Identify means and extremes in the following:

(i) 2 : 3 = 10 : 15

Solution:

Here

Means are 3 and 10

Extremes are 2 and 15

(ii) 5 : 1 = 10 : 2

Solution:

Here

Means are 1 and 10

Extremes are 5 and 2

Example 2: Decide whether the following numbers are in proportion or not.

(i) 2, 3, 4, 6 **(ii)** 5, 10, 15, 20

Solution:

(i) 2, 3, 4, 6

Here

2 : 3 = 4 : 6

Here Product of means = 3 × 4 = 12

and product of extremes = 2 × 6 = 12

Product of means = Product of extremes

2, 3, 4, 6 are in proportion

(ii) 5 : 10 = 15 : 20

Product of means = 10 × 15 = 150

Product of extremes = 5 × 20 = 100

Product of means ≠ Product of extremes

5, 10, 15, 20 are not in proportion

Example 3: Find the value of unknowns in the following proportions:

(i) 33 : 3 :: x : 27 **(ii)** 9 : 6 :: 6 : y

Solution:

(i) 33 : 3 :: x : 27

(product of extremes = product of means)

or $33 \times 27 = 3 \times x$

or $\dfrac{33 \times \cancel{27}^{9}}{\cancel{3}} = x$

or $x = 33 \times 9$

or $x = 297$

(ii) 9 : 6 :: 6 : y

or $9 \times y = 6 \times 6$

or $y = \dfrac{\cancel{6}^{2} \times \cancel{6}^{2}}{\cancel{\cancel{9}_{3}}_{1}}$

or $y = 2 \times 2$

or $y = 4$

Example 4: Find fourth proportional of 2, 5, 8.
Solution: Let fourth proportional be x.

So, we have

2 : 5 = 8 : x

or $2 \times x = 5 \times 8$

or $\dfrac{(2 \times x)}{2} = \dfrac{(5 \times 8)}{2}$

or $x = 20$

So, fourth proportional is 20.

EXERCISE 6.2

Q1. Identify means and extremes in the following:

(I) 2 : 5 = 8 : 20 **(ii)** 3 : 4 = 6 : 8 **(iii)** a : b = c : d

Q2. Decide whether the four numbers given in each of the following are in proportion or not:

(i) 18, 24, 30 and 40
(ii) 14, 19, 3 and 4
(iii) 6, 8, 12 and 16
(iv) 15, 20, 16 and 21
(v) 20, 30, 40 and 50
(vi) 21, 57, 28 and 76

Q3. Find the fourth proportional in the following:

(i) 2, 3 and 6
(ii) $\frac{11}{24}$, $\frac{7}{15}$ and $\frac{5}{8}$
(iii) 16, 12 and 8
(iv) 76, 28 and 57
(v) 36, 45 and 4
(vi) 40, 30 and 24

(1) Direct proportion:

When two quantities are related in such a way that if one quantity increases in a given ratio, the other also increases in the same ratio (and vice versa). This is called **Direct Proportion**.

Examples:

(i) More construction, more materials
(ii) Less money, less shopping
(iii) More students, more teachers
(iv) Less workers, less work

(2) Inverse Proportion:

When two quantities are related in such a way that if one quantity increases in a given ratio, the other decreases in the same ratio. This is called **inverse proportion**.

Examples:

(i) Less speed; more time taken

(ii) More speed; less time taken

(iii) More time; less workers required

(iv) Less time; more workers required

Real life problems involving direct and inverse proportion:

Example 1: If 20 metre cloth is used to make 4 dresses. How much cloth is required to make such 15 dresses?

Solution: Suppose the required cloth is x metre:

Dresses	**Cloth in metres**
4	20
15	x

(More cloth more dresses)

Direct Proportion

$$4 : 15 :: 20 : x$$

As product of means = Product of extremes

or $4 \times x = 15 \times 20$

or $x = \dfrac{15 \times \cancel{20}^{5}}{\cancel{4}_{1}}$

or $x = 75$

Hence the required cloth is 75 metre.

Example 2: If 3 pipes can fill a tank in 80 minutes. How long will it take to fill the tank if 5 pipes are used?

Solution: Let x minutes are required.

Pipes	**Time in minutes**
3	80
5	x

(More time less pipes)

Inverse Proportion

$5 : 3 :: 80 : x$

As product of means = Product of extremes

or $x \times 5 = 3 \times 80$

or $x = \frac{3 \times \cancel{80}^{16}}{\cancel{5}_{1}}$

or $x = 48$

Hence 48 minutes are required.

WORD PROBLEMS

1. Ahmed secured 60 marks in a paper of 100 marks. What shall be his score if the paper were of 75 marks?

2. The weight $\frac{5}{9}$ of a piece of metal is 35 kg. What is the weight $\frac{2}{7}$ of the piece?

3. On a map, 80 km are represented by 5 cm. If the distance between two cities on the map is 15 cm. Find the actual distance between them.

4. If a dozen eggs costs Rs 60, find the cost of 32 eggs.

5. Tabassum earns Rs. 42,000 in a month and spends Rs 39,800. Find the ratio of her savings to her income. Also find the ratio of her expenditure to her income.

6. If Naveed can read 45 pages of a book in 75 minutes, find the time he takes to read the book of 876 pages.

7. If 5 workers can complete a task in 20 days, how many days will 10 workers take to complete the same task?

8. A car travels 120 kilometres in 2 hours. How long will it take to travel 180 kilometres at the same speed?

9. If a 10-litre tank is filled in 4 hours, how long will it take to fill a 40-litre tank at the same rate?

10. A person can complete a task in 30 minutes. If the task is done by 3 people, how long will it take them to finish the task?